# General Physics

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## 1 Derivation

1D-wave Eqn.

The LWR model is a continuous continuity equation,

$$\begin{cases} \frac{\partial \rho}{\partial t} + \frac{\partial q}{\partial x} = 0, & t > 0, \\ \rho(x,0) = \rho_0(x), & t = 0. \end{cases} \tag{1}$$

where $\rho(x,t)$ is the traffic density, $q(x,t) = \rho v(\rho)$ is the traffic flux.

This can be discretised to the CTM model, dividing the road into discrete cells and using discrete time steps giving the following update equation

flow in

$$\Delta\rho_i \rightarrow \quad \rho_i(t + \Delta t) - \rho_i(t) = \frac{\Delta t}{\Delta x}[y_{i-1}(t) - y_i(t)] \tag{2}$$

flow out

where $y_i(t)$ is the flux from cell $i$ to $i+1$ at time $t$, $\Delta x$ is the width of the cell, $\Delta t$ is the time jump and is defined as

Sets a limiter on flux.

$$y_i \quad y_i(t) = \min[D_i(t), S_{i+1}(t)] \tag{3}$$

min of the two

where $D$ and $S$ are previous cell's desire to send vehicles and the current cell's ability to accept vehicles:

$D_i$

$S_{i+1}$

$$D_i(t) = \min[q(\rho_i(t)), q_{max}] \tag{4}$$

min of current flux or max flux.

$$S_{i+1}(t) = \min[q_{max}, w(\rho_j - \rho_{i+1}(t))] \tag{5}$$

$w \cdot (\rho_j - \rho_{i+1})$

where $\rho_j$ is the maximum density, or the jam density.

min of max flux or available flux.

$D$ is essentially how many cars want to leave the upstream cell, and so is just the demand of the current cell.

$S$ is the supply, or room available in the next cell and so is the possible flux into the next cell.

- not a fine.

$w(\rho_j - \rho_{i+1})$ is a measure of how many vehicles can be accepted by the next cell per unit time. - back-propagation of congestion from down to upstream and

- multiplied together

$w \cdot (\rho_i - \rho_{i+1})$ gives a flux which is high for low congestion.

w is the wave speed. The update rule back-propagates upstream slowing down the traffic.

Therefore $y_i(t)$ is the flux into the next cell, determined by whichever is lower, the ability of the next cell to take in cars or the ability of the previous cell to provide them.

# 2 Methodology

## 2.1 Initialisation and Stepping

Initialise the system with $\rho_i(t) = \rho_0(x_i)$ for each cell. Then also initialise $D_i(0)$, $S_{i+1}(0)$ and from these compute $y_i(0)$.

Step forward once in time and compute $D$,$S$ and $y$ and then update $\rho_i$ using Equation 2.

## 2.2 Velocity

The discrete velocity field is defined as

$$v_i(t) = \frac{y_i(t)}{\rho_i(t)} \tag{6}$$

## 2.3 Travel Time

We can then determine the time it took for a vehicle to travel between two positions using

$$T = \sum_{t=0}^{t_{\text{end}}} \Delta t, \quad \text{where } x_v(t_{\text{end}}) \leq x_{\text{end}} \tag{7}$$

which can be determined by updating a vehicles position with

$$x_v(t + \Delta t) = x_v(t) + v_i(t)\Delta t \tag{8}$$

and counting up $\Delta t$.

$$T = \min\left\{ t \,\middle|\, x_v(t) \geq x_{\text{end}}, \quad x_v(t + \Delta t) = x_v(t) + v_{i(t)}\Delta t \right\} \tag{9}$$